# AI, ML & MLOps Interview Preparation Quiz – Answer Sheet

## sundarapandian23@gmail.com

Score: **180 / 250** (72.0%) · Grade: **B** · Rank: **16 of 29** · Correct: 180 · Wrong: 70 · Not answered: 0

Grade bands: A+ 90%+ · A 80–89% · B 70–79% · C 60–69% · D 50–59% · below 50% Needs improvement

### Topic-wise score

| Topic | Score | % |
|---|---|---|
| Machine Learning Fundamentals | 19 / 22 | 86% |
| Deep Learning | 14 / 18 | 78% |
| Large Language Models (LLMs) | 18 / 23 | 78% |
| Prompt Engineering | 11 / 15 | 73% |
| Retrieval-Augmented Generation (RAG) | 15 / 20 | 75% |
| Advanced & Multi-RAG | 1 / 10 | 10% |
| Harness Engineering & AI Agents | 3 / 10 | 30% |
| Model Context Protocol (MCP) & MCP Inspector | 12 / 20 | 60% |
| Ollama & Local LLMs | 8 / 12 | 67% |
| MLOps & MLflow | 15 / 20 | 75% |
| Streamlit | 6 / 10 | 60% |
| Docker | 11 / 15 | 73% |
| Kubernetes (K8s) | 11 / 15 | 73% |
| Git & GitHub | 11 / 12 | 92% |
| APIs & FastAPI | 11 / 12 | 92% |
| Deployment (Render) | 5 / 6 | 83% |
| Jenkins & CI/CD | 9 / 10 | 90% |

Topics shaded red are below 60% – revise these first.

### Question-by-question review (✓ = correct, ✗ = wrong, – = not answered)

| # | Question | Your answer | Correct answer | | Explanation |
|---|---|---|---|---|---|
| **Machine Learning Fundamentals** | | | | | |
| 1 | Supervised learning requires which type of data? | D) Labeled training data | D) Labeled training data | ✓ | Supervised learning learns a mapping from inputs to known labels/targets. |
| 2 | Which of these is a classification algorithm? | A) Logistic Regression | A) Logistic Regression | ✓ | Despite its name, logistic regression predicts class probabilities. |
| 3 | K-Means is best described as: | A) An unsupervised clustering algorithm | A) An unsupervised clustering algorithm | ✓ | K-Means groups unlabeled points into k clusters by distance to centroids. |
| 4 | A model performs very well on training data but poorly on test data. This is: | C) Overfitting | C) Overfitting | ✓ | Overfitting means the model memorized noise in training data and fails to generalize. |
| 5 | High bias in a model typically leads to: | D) Underfitting | D) Underfitting | ✓ | High bias = overly simple assumptions, so the model misses real patterns. |
| 6 | Which technique helps reduce overfitting? | C) L1/L2 regularization | C) L1/L2 regularization | ✓ | Regularization penalizes large weights, discouraging overly complex models. |
| 7 | L1 regularization (Lasso) tends to: | C) Drive some weights to exactly zero, performing feature selection | C) Drive some weights to exactly zero, performing feature selection | ✓ | The L1 penalty produces sparse solutions, effectively dropping features. |
| 8 | Precision is calculated as: | A) TP / (TP + FP) | A) TP / (TP + FP) | ✓ | Precision = of everything predicted positive, how many were actually positive. |
| 9 | Recall (sensitivity) is calculated as: | A) TP / (TP + FP) | B) TP / (TP + FN) | ✗ | Recall = of all actual positives, how many the model found. |
| 10 | For a fraud dataset where only 1% of transactions are fraud, which metric is most useful? | C) F1-score or PR-AUC | C) F1-score or PR-AUC | ✓ | Accuracy is misleading on imbalanced data: predicting 'not fraud' always gives 99%. |
| 11 | The F1-score is: | B) The harmonic mean of precision and recall | B) The harmonic mean of precision and recall | ✓ | F1 = 2PR/(P+R); it is high only when both precision and recall are high. |

| # | Question | Your answer | Correct answer | | Explanation |
|---|---|---|---|---|---|
| 12 | What is the main purpose of k-fold cross-validation? | C) To estimate model performance on unseen data more reliably | C) To estimate model performance on unseen data more reliably | ✓ | Each fold serves once as validation, giving a more stable performance estimate. |
| 13 | PCA (Principal Component Analysis) is mainly used for: | A) Dimensionality reduction | A) Dimensionality reduction | ✓ | PCA projects data onto directions of maximum variance, reducing feature count. |
| 14 | Data leakage in ML refers to: | D) Information from outside the training data (e.g., the target or test set) influencing training, inflating performance | D) Information from outside the training data (e.g., the target or test set) influencing training, inflating performance | ✓ | Leakage gives unrealistically good validation scores that collapse in production. |
| 15 | Gradient descent works by: | D) Iteratively updating parameters in the direction of the negative gradient to minimize loss | D) Iteratively updating parameters in the direction of the negative gradient to minimize loss | ✓ | Moving opposite to the gradient reduces the loss step by step. |
| 16 | If the learning rate is set too high, what typically happens? | B) The model always converges faster to the optimum | A) The loss oscillates or diverges | ✗ | Steps that are too large overshoot the minimum. |
| 17 | Random Forest is: | A) An ensemble of decision trees trained with bagging | A) An ensemble of decision trees trained with bagging | ✓ | Random Forest averages many trees built on bootstrap samples with random feature subsets. |
| 18 | XGBoost is based on which technique? | D) Gradient boosting | D) Gradient boosting | ✓ | XGBoost builds trees sequentially, each correcting errors of the previous ones. |
| 19 | Feature scaling (standardization) is most important for: | D) Decision trees | C) Distance- or gradient-based algorithms like KNN, SVM and neural networks | ✗ | Unscaled features with large ranges dominate distances and slow gradient descent. |
| 20 | One-hot encoding is used to: | B) Convert categorical variables into binary vectors | B) Convert categorical variables into binary vectors | ✓ | Each category becomes its own 0/1 column. |
| 21 | ROC-AUC measures: | C) How well a model ranks positives above negatives across all thresholds | C) How well a model ranks positives above negatives across all thresholds | ✓ | AUC of 0.5 is random ranking; 1.0 is perfect separation. |
| 22 | The bias-variance tradeoff states that: | A) Reducing bias often increases variance and vice versa | A) Reducing bias often increases variance and vice versa | ✓ | Model complexity balances underfitting (bias) against overfitting (variance). |
| **Deep Learning** | | | | | |
| 23 | Why do neural networks use activation functions? | B) To normalize the input data | A) To introduce non-linearity | ✗ | Without non-linear activations, stacked layers collapse into one linear transformation. |
| 24 | The ReLU activation function is defined as: | A) max(0, x) | A) max(0, x) | ✓ | ReLU outputs x if positive, else 0. |
| 25 | The vanishing gradient problem is most associated with: | B) Sigmoid/tanh activations in very deep networks | B) Sigmoid/tanh activations in very deep networks | ✓ | Sigmoid/tanh derivatives are small, so gradients shrink as they are multiplied back through layers. |
| 26 | Backpropagation is used to: | B) Compute gradients of the loss with respect to weights using the chain rule | B) Compute gradients of the loss with respect to weights using the chain rule | ✓ | Backprop propagates error from output to input layers to compute gradients. |
| 27 | Dropout is a technique that: | D) Randomly deactivates neurons during training to reduce overfitting | D) Randomly deactivates neurons during training to reduce overfitting | ✓ | Dropout forces the network not to rely on any single neuron; it is disabled at inference. |
| 28 | Batch Normalization helps by: | B) Normalizing layer inputs over a mini-batch to stabilize and speed up training | B) Normalizing layer inputs over a mini-batch to stabilize and speed up training | ✓ | BatchNorm keeps activations in a stable range, allowing higher learning rates. |
| 29 | Convolutional Neural Networks (CNNs) are primarily used for: | A) Image and grid-like data | A) Image and grid-like data | ✓ | Convolutions capture local spatial patterns such as edges and textures. |
| 30 | What does a pooling layer in a CNN do? | A) Downsamples feature maps to reduce spatial size | A) Downsamples feature maps to reduce spatial size | ✓ | Max/average pooling reduces size and computation while keeping key features. |
| 31 | LSTM networks were introduced mainly to address: | A) Overfitting in CNNs | B) Vanishing gradients and long-term dependencies in RNNs | ✗ | LSTM gates control information flow, letting it remember over long sequences. |

| # | Question | Your answer | Correct answer | | Explanation |
|---|---|---|---|---|---|
| 32 | The key mechanism of the Transformer architecture is: | A) Self-attention | A) Self-attention | ✓ | Self-attention lets every token weigh every other token in parallel ('Attention Is All You Need'). |
| 33 | The softmax function outputs: | D) A probability distribution that sums to 1 | D) A probability distribution that sums to 1 | ✓ | Softmax converts logits into class probabilities. |
| 34 | Cross-entropy loss is typically used for: | B) Regression tasks | C) Classification tasks | ✗ | It measures the gap between predicted probabilities and the true class. |
| 35 | One epoch in training means: | C) One complete pass over the entire training dataset | C) One complete pass over the entire training dataset | ✓ | Number of steps per epoch = dataset size / batch size. |
| 36 | Transfer learning means: | B) Reusing a model pretrained on one task as a starting point for a new task | B) Reusing a model pretrained on one task as a starting point for a new task | ✓ | E.g., fine-tuning an ImageNet-trained ResNet on medical images. |
| 37 | The Adam optimizer: | A) Combines momentum with adaptive per-parameter learning rates | A) Combines momentum with adaptive per-parameter learning rates | ✓ | Adam = momentum + RMSProp-style adaptive scaling. |
| 38 | Early stopping means: | B) Stopping training when validation performance stops improving | B) Stopping training when validation performance stops improving | ✓ | It prevents overfitting by halting before validation loss rises. |
| 39 | An embedding layer: | B) Creates sparse one-hot vectors | D) Maps discrete tokens/IDs to dense, learned vectors | ✗ | Embeddings place similar items close together in vector space. |
| 40 | Why are GPUs preferred for deep learning? | D) They perform massively parallel matrix operations | D) They perform massively parallel matrix operations | ✓ | Neural network training is dominated by matrix multiplications that parallelize well. |
| **Large Language Models (LLMs)** | | | | | |
| 41 | What does LLM stand for? | C) Large Language Model | C) Large Language Model | ✓ | LLMs are transformer models trained on huge text corpora. |
| 42 | Tokenization in LLMs means: | A) Splitting text into tokens (often subwords) the model can process | A) Splitting text into tokens (often subwords) the model can process | ✓ | Methods like BPE break words into subword units mapped to IDs. |
| 43 | The 'context window' of an LLM is: | C) The maximum number of tokens the model can consider in one request (input + output) | C) The maximum number of tokens the model can consider in one request (input + output) | ✓ | Anything beyond the context window cannot be seen by the model. |
| 44 | The temperature parameter controls: | B) The randomness of the generated output | B) The randomness of the generated output | ✓ | Lower temperature = more focused; higher = more varied/creative. |
| 45 | Setting temperature close to 0 makes outputs: | B) More deterministic, favoring the most likely tokens | B) More deterministic, favoring the most likely tokens | ✓ | Near-zero temperature is good for factual or structured tasks. |
| 46 | Top-p (nucleus) sampling: | C) Samples from the smallest set of tokens whose cumulative probability reaches p | C) Samples from the smallest set of tokens whose cumulative probability reaches p | ✓ | Top-p adapts the candidate pool size to the model's confidence. |
| 47 | An LLM 'hallucination' is: | C) Fluent output that is factually incorrect or fabricated | C) Fluent output that is factually incorrect or fabricated | ✓ | Mitigations include RAG grounding, citations and lower temperature. |
| 48 | Which model family uses a decoder-only architecture? | B) GPT | B) GPT | ✓ | GPT models generate text autoregressively with a decoder-only transformer. |
| 49 | BERT is primarily: | B) An encoder-only model pretrained with masked language modeling | B) An encoder-only model pretrained with masked language modeling | ✓ | BERT excels at understanding tasks like classification and NER. |
| 50 | Fine-tuning an LLM means: | D) Further training a pretrained model on task-specific data | D) Further training a pretrained model on task-specific data | ✓ | Fine-tuning adapts behavior, style or domain knowledge. |
| 51 | LoRA (Low-Rank Adaptation) is: | D) A parameter-efficient fine-tuning method that trains small low-rank adapter matrices while freezing base weights | D) A parameter-efficient fine-tuning method that trains small low-rank adapter matrices while freezing base weights | ✓ | LoRA cuts memory and compute; QLoRA combines it with 4-bit quantization. |

| # | Question | Your answer | Correct answer | | Explanation |
|---|---|---|---|---|---|
| 52 | Model quantization means: | D) Reducing the numerical precision of weights (e.g., FP16 to INT4) to save memory and speed up inference | D) Reducing the numerical precision of weights (e.g., FP16 to INT4) to save memory and speed up inference | ✓ | Quantized models (like GGUF Q4) run on laptops with small quality loss. |
| 53 | RLHF stands for: | D) Regression Learning for Hidden Features | A) Reinforcement Learning from Human Feedback | ✗ | Humans rank outputs, a reward model learns preferences, and the LLM is optimized against it. |
| 54 | In LLM applications, an 'embedding' is: | C) A compressed text file | B) A numeric vector capturing the semantic meaning of text | ✗ | Similar meanings produce nearby vectors, enabling semantic search. |
| 55 | Cosine similarity measures: | B) The similarity of two vectors based on the angle between them | B) The similarity of two vectors based on the angle between them | ✓ | Values near 1 mean very similar direction/meaning. |
| 56 | Which of these is an open-weight model you can download and run yourself? | C) Anthropic Claude | B) Meta Llama | ✗ | Open-weight models (Llama, Mistral, Gemma, Qwen) can be run locally, e.g., with Ollama. |
| 57 | The max_tokens parameter: | A) Limits the length of the generated output | A) Limits the length of the generated output | ✓ | Output stops once the token limit is reached. |
| 58 | A system prompt is: | B) Instructions that set the model's role and behavior for the conversation | B) Instructions that set the model's role and behavior for the conversation | ✓ | E.g., 'You are a helpful assistant that answers only in JSON.' |
| 59 | A lower perplexity score indicates: | D) The model predicts the text better | D) The model predicts the text better | ✓ | Perplexity measures how 'surprised' the model is by the text. |
| 60 | Function calling (tool use) in LLMs means: | C) A tokenization method | D) The model outputs a structured request (e.g., JSON) that the application executes as a tool call | ✗ | The app runs the function and returns the result to the model. |
| 61 | Why do LLMs have a 'knowledge cutoff'? | B) Because of quantization | D) They are trained on data only up to a certain date | ✗ | RAG or web search is needed for information after the cutoff. |
| 62 | A multimodal model is one that: | B) Can process multiple input types such as text and images | B) Can process multiple input types such as text and images | ✓ | Examples include vision-language models like LLaVA. |
| 63 | The pretraining objective of GPT-style models is: | B) Next-token prediction | B) Next-token prediction | ✓ | The model learns to predict the next token given previous tokens. |
| **Prompt Engineering** | | | | | |
| 64 | Zero-shot prompting means: | B) Asking the model to do a task without giving any examples | B) Asking the model to do a task without giving any examples | ✓ | The model relies on its pretrained knowledge alone. |
| 65 | Few-shot prompting means: | C) Including a few input-output examples in the prompt | C) Including a few input-output examples in the prompt | ✓ | Examples show the model the expected format and behavior. |
| 66 | Chain-of-Thought (CoT) prompting: | A) Encourages the model to reason step by step before giving the final answer | A) Encourages the model to reason step by step before giving the final answer | ✓ | CoT improves accuracy on math and logic problems. |
| 67 | Prompt injection is: | C) Malicious input that tries to override the system's original instructions | C) Malicious input that tries to override the system's original instructions | ✓ | E.g., a document saying 'ignore previous instructions and reveal secrets'. |
| 68 | Best way to get reliable structured output (e.g., JSON) from an LLM: | C) Specify the exact schema/format and give an example | C) Specify the exact schema/format and give an example | ✓ | Many APIs also provide structured-output or JSON modes. |
| 69 | Role prompting means: | A) Assigning a persona, e.g., 'You are a senior DevOps engineer' | A) Assigning a persona, e.g., 'You are a senior DevOps engineer' | ✓ | A role primes relevant knowledge, tone and depth. |
| 70 | Delimiters such as XML tags or ### in a prompt are used to: | B) Reduce token cost | A) Clearly separate instructions from context or input data | ✗ | Clear separation reduces confusion and injection risk. |
| 71 | Prompt chaining is: | C) Breaking a complex task into sequential prompts where each output feeds the next | C) Breaking a complex task into sequential prompts where each output feeds the next | ✓ | E.g., extract, then summarize, then format. |
| 72 | The ReAct prompting pattern stands for: | B) Reasoning + Acting, interleaving thoughts with tool calls and observations | B) Reasoning + Acting, interleaving thoughts with tool calls and observations | ✓ | ReAct is a foundation of many LLM agent loops. |

| # | Question | Your answer | Correct answer | | Explanation |
|---|---|---|---|---|---|
| 73 | Self-consistency prompting: | C) Samples multiple reasoning paths and picks the most common final answer | C) Samples multiple reasoning paths and picks the most common final answer | ✓ | Majority voting across samples improves reliability. |
| 74 | Which prompt strategy helps reduce hallucinations? | B) Remove all context | C) Provide grounding context and allow the model to say 'I don't know' | ✗ | Grounding plus permission to abstain discourages fabrication. |
| 75 | A prompt template is: | B) A fixed prompt that cannot change | A) A reusable prompt with placeholders for variables | ✗ | E.g., 'Summarize the following {document} in {n} bullets.' |
| 76 | Instead of only telling the model what NOT to do, it is usually better to: | B) Clearly tell it what TO do with specific, positive instructions | B) Clearly tell it what TO do with specific, positive instructions | ✓ | Positive, specific instructions are easier for models to follow. |
| 77 | Why evaluate prompts against a test set? | D) To measure quality changes objectively across prompt versions | D) To measure quality changes objectively across prompt versions | ✓ | Prompt changes can fix one case while breaking others. |
| 78 | Tree of Thoughts (ToT) prompting: | D) Stores prompts in a folder tree | B) Explores and evaluates multiple reasoning branches before choosing | ✗ | It generalizes CoT into a search over possible thoughts. |
| **Retrieval-Augmented Generation (RAG)** | | | | | |
| 79 | RAG stands for: | D) Retrieval-Augmented Generation | D) Retrieval-Augmented Generation | ✓ | RAG retrieves relevant documents and passes them to the LLM as context. |
| 80 | The main benefit of RAG is: | A) Grounding answers in external, up-to-date data without retraining the model | A) Grounding answers in external, up-to-date data without retraining the model | ✓ | RAG also enables citations and private-data Q&A. |
| 81 | The correct order of a typical RAG pipeline is: | B) Load, chunk, embed, store in vector DB, retrieve, generate | B) Load, chunk, embed, store in vector DB, retrieve, generate | ✓ | Indexing happens offline; retrieval and generation happen per query. |
| 82 | Why do we chunk documents in RAG? | D) To create smaller pieces that fit the context window and improve retrieval precision | D) To create smaller pieces that fit the context window and improve retrieval precision | ✓ | Chunk size is a key tuning parameter. |
| 83 | What is the purpose of chunk overlap? | C) To preserve context across chunk boundaries | C) To preserve context across chunk boundaries | ✓ | Overlap prevents sentences from being cut off from their context. |
| 84 | Which of these is a vector database? | D) Chroma | D) Chroma | ✓ | Others include FAISS, Pinecone, Qdrant, Weaviate, Milvus and pgvector. |
| 85 | FAISS is: | D) A library from Meta for efficient similarity search over dense vectors | D) A library from Meta for efficient similarity search over dense vectors | ✓ | FAISS supports exact and approximate nearest-neighbor indexes. |
| 86 | Top-k retrieval means: | B) Returning the k most similar chunks to the query | B) Returning the k most similar chunks to the query | ✓ | k trades recall against context size and noise. |
| 87 | Hybrid search combines: | C) Two different LLMs | B) Keyword search (e.g., BM25) with semantic vector search | ✗ | Keywords catch exact terms; vectors catch meaning. |
| 88 | BM25 is a: | B) Keyword/lexical ranking function | B) Keyword/lexical ranking function | ✓ | BM25 scores documents by term frequency and inverse document frequency. |
| 89 | A re-ranker in RAG: | B) Re-chunks documents | D) Re-scores retrieved candidates (e.g., with a cross-encoder) for better relevance ordering | ✗ | Retrieve broadly, then re-rank to the best few. |
| 90 | For embeddings in a RAG system you should: | C) Use the same embedding model for indexing and querying | C) Use the same embedding model for indexing and querying | ✓ | Vectors from different models live in different spaces and cannot be compared. |
| 91 | Metadata filtering in retrieval allows you to: | D) Restrict results using attributes like date, source or department | D) Restrict results using attributes like date, source or department | ✓ | E.g., only search HR documents from 2025. |
| 92 | Your RAG system keeps retrieving irrelevant chunks. A good first fix is: | C) Tune chunking and retrieval (chunk size, hybrid search, re-ranking) | C) Tune chunking and retrieval (chunk size, hybrid search, re-ranking) | ✓ | Most RAG quality problems are retrieval problems. |
| 93 | RAGAS is used for: | B) Tokenizing documents | A) Evaluating RAG pipelines with metrics like faithfulness and context precision | ✗ | It also measures answer relevancy and context recall. |

| # | Question | Your answer | Correct answer | | Explanation |
|---|---|---|---|---|---|
| 94 | The 'faithfulness' metric in RAG evaluation checks: | D) Whether the answer is supported by the retrieved context | D) Whether the answer is supported by the retrieved context | ✓ | Low faithfulness indicates hallucination beyond the context. |
| 95 | The 'lost in the middle' problem means: | A) LLMs may underuse information placed in the middle of long contexts | A) LLMs may underuse information placed in the middle of long contexts | ✓ | Place the most relevant chunks at the start or end of the context. |
| 96 | If your knowledge base changes daily, the better approach is: | B) RAG | B) RAG | ✓ | RAG updates knowledge by re-indexing documents, not retraining. |
| 97 | Approximate Nearest Neighbor (ANN) indexes like HNSW: | D) Only work with keyword search | A) Trade a little accuracy for much faster vector search | ✗ | HNSW builds a navigable graph for sub-linear search. |
| 98 | LangChain and LlamaIndex are: | D) GPU drivers | A) Frameworks for building LLM applications (loaders, splitters, retrievers, chains) | ✗ | They speed up building RAG and agent pipelines. |
| **Advanced & Multi-RAG** | | | | | |
| 99 | Multimodal RAG refers to: | D) Retrieval over multiple data types such as text, images and tables | D) Retrieval over multiple data types such as text, images and tables | ✓ | It often uses vision models or captions for images and tables. |
| 100 | Multi-query retrieval: | C) Sends the same query to the same index many times | D) Uses an LLM to generate several reformulations of the query to improve recall | ✗ | Results from all variants are merged and deduplicated. |
| 101 | HyDE (Hypothetical Document Embeddings): | C) Encrypts embeddings | B) Generates a hypothetical answer, embeds it, and uses it to retrieve real documents | ✗ | A fake answer is often closer in vector space to real answers than the question is. |
| 102 | Agentic RAG means: | B) A RAG system with fixed retrieval only | A) An agent decides when, where and how to retrieve, possibly iteratively using tools | ✗ | The agent can rewrite queries, pick sources and verify answers. |
| 103 | In a multi-source RAG system, 'routing' means: | C) Load balancing GPUs | D) Directing the query to the most relevant data source or index | ✗ | E.g., HR questions go to the HR index, code questions to the code index. |
| 104 | GraphRAG uses: | D) Bar charts to display results | B) A knowledge graph of entities and relationships to support multi-hop reasoning | ✗ | Graphs help answer questions that connect facts across documents. |
| 105 | Parent-document retrieval: | B) Retrieves documents created by parents | D) Matches on small chunks but returns the larger parent section as context | ✗ | Small chunks match precisely; large parents give full context. |
| 106 | Corrective RAG (CRAG): | C) Always skips retrieval | D) Grades retrieved documents and, if they are poor, takes corrective action such as a web search | ✗ | It adds a self-check step on retrieval quality. |
| 107 | Reciprocal Rank Fusion (RRF) is used to: | C) Encrypt queries | A) Merge ranked results from multiple retrievers into one list | ✗ | Common in hybrid search to combine BM25 and vector results. |
| 108 | A 'multi-hop' question in RAG is one that: | C) Contains multiple spelling errors | D) Requires combining information from multiple documents to answer | ✗ | E.g., 'Who manages the team that built product X?' |
| **Harness Engineering & AI Agents** | | | | | |
| 109 | In AI, 'harness engineering' refers to: | B) Manually labeling training data | A) Designing the scaffolding around an LLM (tools, context management, control loop, guardrails, evals) to make it a reliable agent | ✗ | The same model can perform very differently depending on its harness. |
| 110 | The core loop of an LLM agent is: | B) Train, test, deploy, and stop | C) Reason, call a tool, observe the result, and repeat until the task is done | ✗ | This loop is how agents complete multi-step tasks. |
| 111 | Context management in long-running agents involves: | C) Summarizing or compacting history and selecting relevant information to stay within the context window | C) Summarizing or compacting history and selecting relevant information to stay within the context window | ✓ | Good context management prevents agents from losing track or running out of tokens. |

| # | Question | Your answer | Correct answer | | Explanation |
|---|---|---|---|---|---|
| 112 | Guardrails in an AI agent system are: | B) Physical barriers around servers | D) Checks on inputs, outputs and actions that keep the agent safe and on-policy | ✗ | Examples: blocking PII leaks, validating tool arguments, content filters. |
| 113 | Human-in-the-loop for agents means: | B) Users must solve a CAPTCHA each turn | C) Requiring human approval before high-risk actions | ✗ | E.g., confirming before sending emails, payments or deleting data. |
| 114 | An evaluation harness (e.g., lm-evaluation-harness) is: | A) A Python web framework | D) A framework that runs models or agents against benchmark tasks and scores them automatically | ✗ | It makes performance comparisons repeatable. |
| 115 | Why set a maximum number of steps/iterations for an agent? | A) To prevent infinite loops and runaway cost | A) To prevent infinite loops and runaway cost | ✓ | Budgets on steps, time and tokens are standard harness controls. |
| 116 | Why do tool names, descriptions and schemas matter so much in agents? | B) They control GPU allocation | C) The model decides which tool to use and how to call it based on them | ✗ | Clear, specific tool descriptions greatly improve tool-use accuracy. |
| 117 | In multi-agent orchestration, a supervisor/orchestrator agent: | B) Delegates subtasks to specialized agents and combines results | B) Delegates subtasks to specialized agents and combines results | ✓ | Frameworks like LangGraph and CrewAI support this pattern. |
| 118 | Tools like LangSmith or Langfuse provide agent observability by: | C) Hosting the model weights | A) Tracing each step, tool call, token usage and latency for debugging | ✗ | Tracing is essential for debugging non-deterministic agent behavior. |
| **Model Context Protocol (MCP) & MCP Inspector** | | | | | |
| 119 | MCP stands for: | D) Model Context Protocol | D) Model Context Protocol | ✓ | MCP standardizes how AI apps connect to tools and data. |
| 120 | Which company introduced MCP as an open standard? | B) Anthropic | B) Anthropic | ✓ | Anthropic open-sourced MCP in November 2024; it is now widely adopted. |
| 121 | The main purpose of MCP is to: | B) Provide an open standard for connecting AI applications to external tools and data sources | B) Provide an open standard for connecting AI applications to external tools and data sources | ✓ | Often described as 'USB-C for AI apps'. |
| 122 | The three core primitives an MCP server can expose are: | A) Tools, Resources and Prompts | A) Tools, Resources and Prompts | ✓ | Tools = actions, Resources = data, Prompts = templates. |
| 123 | In MCP, 'Tools' are: | A) Functions the model can invoke to perform actions | A) Functions the model can invoke to perform actions | ✓ | E.g., create_ticket, query_database, send_email. |
| 124 | In MCP, 'Resources' are: | B) Executable shell commands only | C) Data or content (files, records) exposed as context, identified by URIs | ✗ | Resources are application-controlled context, like a file's contents. |
| 125 | In MCP, 'Prompts' are: | A) Reusable prompt templates/workflows exposed by the server, typically chosen by the user | A) Reusable prompt templates/workflows exposed by the server, typically chosen by the user | ✓ | They often appear as slash commands in clients. |
| 126 | MCP messages use which format? | D) Protocol Buffers only | A) JSON-RPC 2.0 | ✗ | Requests, responses and notifications follow JSON-RPC 2.0. |
| 127 | The standard MCP transports are: | A) Only gRPC | C) stdio and Streamable HTTP | ✗ | Streamable HTTP replaced the older HTTP+SSE transport. |
| 128 | The stdio transport is typically used when: | B) The MCP server runs as a local subprocess of the client | B) The MCP server runs as a local subprocess of the client | ✓ | The client communicates over the process's stdin/stdout. |
| 129 | In MCP architecture, each MCP client maintains: | C) A direct connection to model weights | A) A 1:1 connection with a single MCP server | ✗ | The host app (e.g., Claude Desktop or an IDE) creates one client per server. |
| 130 | The MCP Inspector is: | C) A code linter for Python | D) An interactive developer tool for testing and debugging MCP servers | ✗ | You can list and call tools, read resources and test prompts. |
| 131 | How do you typically launch the MCP Inspector? | C) pip install inspector-ai | D) npx @modelcontextprotocol/inspector | ✗ | Append the server's start command, e.g., npx @modelcontextprotocol/inspector python server.py |
| 132 | Which helper in the official Python SDK makes building MCP servers quick? | B) Flask-Login | C) FastMCP | ✗ | from mcp.server.fastmcp import FastMCP |

| # | Question | Your answer | Correct answer | | Explanation |
|---|---|---|---|---|---|
| 133 | In FastMCP, how do you register a Python function as a tool? | B) Put it in a Dockerfile | D) Decorate it with @mcp.tool() | ✗ | Type hints and the docstring become the tool's schema and description. |
| 134 | Capability negotiation between MCP client and server happens during: | B) The initialization handshake (initialize request) | B) The initialization handshake (initialize request) | ✓ | Both sides declare supported features and protocol version. |
| 135 | MCP 'sampling' allows: | B) A server to request an LLM completion through the client | B) A server to request an LLM completion through the client | ✓ | This lets servers use AI without their own model API keys. |
| 136 | Which is a security best practice for MCP servers? | D) Least privilege, input validation and user consent for sensitive tool calls | D) Least privilege, input validation and user consent for sensitive tool calls | ✓ | MCP servers can take real actions, so treat them like any privileged integration. |
| 137 | How does MCP relate to LLM function calling? | A) Function calling is the model-level ability to request a tool; MCP standardizes how tools are discovered and connected across apps | A) Function calling is the model-level ability to request a tool; MCP standardizes how tools are discovered and connected across apps | ✓ | MCP builds on function calling to make tools reusable across hosts. |
| 138 | 'Tool poisoning' in MCP refers to: | C) Malicious instructions hidden in tool descriptions to manipulate the model | C) Malicious instructions hidden in tool descriptions to manipulate the model | ✓ | Only install MCP servers from trusted sources and review their tools. |
| **Ollama & Local LLMs** | | | | | |
| 139 | Ollama is a tool for: | A) Running open LLMs locally on your machine | A) Running open LLMs locally on your machine | ✓ | It packages model weights, config and a local API server. |
| 140 | Which command downloads a model without starting a chat? | C) ollama get llama3 | B) ollama pull llama3 | ✗ | ollama pull downloads the model to your local store. |
| 141 | Which command starts an interactive chat with a model? | A) ollama start llama3 | C) ollama run llama3 | ✗ | ollama run also pulls the model if it is not present. |
| 142 | Which command lists models downloaded locally? | A) ollama list | A) ollama list | ✓ | It shows model names, sizes and modification times. |
| 143 | Ollama's local API server listens on which default port? | B) 11434 | B) 11434 | ✓ | Base URL: http://localhost:11434 |
| 144 | An Ollama Modelfile is used to: | B) Customize a model with a base model (FROM), parameters and a system prompt | B) Customize a model with a base model (FROM), parameters and a system prompt | ✓ | Similar in spirit to a Dockerfile, but for models. |
| 145 | How do you build a custom model from a Modelfile? | C) ollama create mymodel -f Modelfile | C) ollama create mymodel -f Modelfile | ✓ | Then run it with ollama run mymodel. |
| 146 | Which Ollama REST endpoint is used for multi-turn chat with message history? | D) /api/chat | D) /api/chat | ✓ | /api/generate handles single prompt completions. |
| 147 | Models served by Ollama typically use which file format? | C) ONNX only | B) GGUF (quantized) | ✗ | GGUF is the llama.cpp format supporting quantized weights. |
| 148 | A key advantage of running LLMs locally with Ollama is: | B) Data privacy and no per-token API cost | B) Data privacy and no per-token API cost | ✓ | Data never leaves your machine. |
| 149 | To create embeddings for RAG with Ollama you would: | A) Use the Modelfile SYSTEM line | C) Use an embedding model (e.g., nomic-embed-text) via the embed API | ✗ | Embedding models output vectors, not text. |
| 150 | Which command shows which models are currently loaded in memory? | C) ollama ps | C) ollama ps | ✓ | Useful to check GPU/CPU usage of loaded models. |
| **MLOps & MLflow** | | | | | |
| 151 | MLOps is best described as: | A) Practices to deploy, monitor and maintain ML models in production reliably | A) Practices to deploy, monitor and maintain ML models in production reliably | ✓ | It applies DevOps ideas (CI/CD, versioning, monitoring) to ML. |
| 152 | Which are core components of MLflow? | D) Tracking, Projects, Models and Model Registry | D) Tracking, Projects, Models and Model Registry | ✓ | Tracking logs runs; the Registry manages model versions. |
| 153 | MLflow Tracking records: | C) Parameters, metrics, artifacts and code versions of runs | C) Parameters, metrics, artifacts and code versions of runs | ✓ | This makes experiments comparable and reproducible. |
| 154 | How do you log a hyperparameter in MLflow? | C) mlflow.param.set('lr', 0.01) | A) mlflow.log_param('lr', 0.01) | ✗ | Use log_params() to log a dict of many parameters. |
| 155 | How do you log a metric such as accuracy in MLflow? | C) mlflow.log_metric('accuracy', 0.92) | C) mlflow.log_metric('accuracy', 0.92) | ✓ | Metrics can be logged per step to plot training curves. |

| # | Question | Your answer | Correct answer | | Explanation |
|---|---|---|---|---|---|
| 156 | Which statement starts a new MLflow run? | B) mlflow.init_run() | D) with mlflow.start_run(): | ✗ | Everything logged inside the block belongs to that run. |
| 157 | mlflow.autolog() does what? | B) Automatically logs parameters, metrics and models for supported libraries | B) Automatically logs parameters, metrics and models for supported libraries | ✓ | Works with scikit-learn, XGBoost, PyTorch, TensorFlow and more. |
| 158 | How do you open the MLflow tracking UI locally? | A) mlflow ui (default port 5000) | A) mlflow ui (default port 5000) | ✓ | Then browse to http://localhost:5000 |
| 159 | The MLflow Model Registry is used to: | A) Centrally version models and manage their lifecycle (e.g., aliases like 'champion') | A) Centrally version models and manage their lifecycle (e.g., aliases like 'champion') | ✓ | It records lineage from model version back to the training run. |
| 160 | Data drift means: | B) The distribution of production input data changes from the training data | B) The distribution of production input data changes from the training data | ✓ | Drift can silently degrade model accuracy. |
| 161 | Concept drift means: | B) The relationship between inputs and the target changes over time | B) The relationship between inputs and the target changes over time | ✓ | E.g., customer buying behavior changing after a major event. |
| 162 | The main reason for experiment tracking is to: | A) Compare runs and reproduce results | A) Compare runs and reproduce results | ✓ | Without tracking, it is hard to know which settings produced the best model. |
| 163 | DVC (Data Version Control) is used to: | A) Manage Docker containers | C) Version datasets and models alongside Git | ✗ | Large files are stored remotely and referenced by small Git-tracked files. |
| 164 | A feature store (e.g., Feast) provides: | B) A central place to store and serve features consistently for training and inference | B) A central place to store and serve features consistently for training and inference | ✓ | It helps prevent training-serving skew. |
| 165 | Training-serving skew is: | A) A difference between how data is processed in training versus in production serving | A) A difference between how data is processed in training versus in production serving | ✓ | It causes models to underperform in production despite good offline metrics. |
| 166 | A canary deployment of a new model means: | C) Deploying to all users at once | D) Routing a small percentage of traffic to the new model first | ✗ | If metrics look good, traffic is gradually increased. |
| 167 | A/B testing of ML models involves: | B) Splitting live traffic between two models and comparing business/quality metrics | B) Splitting live traffic between two models and comparing business/quality metrics | ✓ | It measures real-world impact, not just offline accuracy. |
| 168 | Production model monitoring should track: | C) Prediction quality, latency and data drift | C) Prediction quality, latency and data drift | ✓ | Model health is more than infrastructure health. |
| 169 | An MLflow model 'flavor' is: | B) A convention (e.g., sklearn, pytorch) describing how to save, load and serve a model | B) A convention (e.g., sklearn, pytorch) describing how to save, load and serve a model | ✓ | The pyfunc flavor offers a generic predict interface. |
| 170 | How can you serve version 1 of a registered MLflow model as a REST API? | B) kubectl serve my_model | A) mlflow models serve -m 'models:/my_model/1' | ✗ | This starts a local scoring server with an /invocations endpoint. |
| **Streamlit** | | | | | |
| 171 | Streamlit is: | B) A Python framework for building data and ML web apps quickly | B) A Python framework for building data and ML web apps quickly | ✓ | Write Python scripts; Streamlit renders the UI. |
| 172 | How do you run a Streamlit app? | A) streamlit run app.py | A) streamlit run app.py | ✓ | The app opens in your browser. |
| 173 | What is Streamlit's default port? | C) 8501 | C) 8501 | ✓ | Change it with --server.port. |
| 174 | Which function creates a single-line text input box? | B) st.textfield() | A) st.text_input() | ✗ | st.text_area() is used for multi-line input. |
| 175 | How does Streamlit's execution model work? | C) The script reruns from top to bottom on every widget interaction | C) The script reruns from top to bottom on every widget interaction | ✓ | This is why caching and session state are important. |
| 176 | To keep values (like chat history) across reruns, use: | C) st.session_state | C) st.session_state | ✓ | session_state persists per user session. |
| 177 | To load an ML model or LLM client once and reuse it across reruns, use: | C) @st.cache_resource | C) @st.cache_resource | ✓ | cache_resource is for unserializable shared objects; cache_data is for data like DataFrames. |

| # | Question | Your answer | Correct answer | | Explanation |
|---|---|---|---|---|---|
| 178 | Which elements are used to build a chat interface in Streamlit? | C) st.chatbox and st.reply | A) st.chat_message and st.chat_input | ✗ | These make LLM chatbot UIs quick to build. |
| 179 | How do users upload a PDF for a RAG app in Streamlit? | B) st.drag_drop() | C) st.file_uploader() | ✗ | You can restrict file types with type=['pdf']. |
| 180 | How do you place widgets in a side panel? | D) st.sidepanel | B) st.sidebar | ✗ | E.g., st.sidebar.selectbox('Model', [...]). |
| **Docker** | | | | | |
| 181 | The main difference between containers and virtual machines is: | B) Containers share the host OS kernel and are lighter; VMs run a full guest OS | B) Containers share the host OS kernel and are lighter; VMs run a full guest OS | ✓ | This makes containers start faster and use less memory. |
| 182 | A Dockerfile is: | A) A text file with instructions to build a Docker image | A) A text file with instructions to build a Docker image | ✓ | Instructions include FROM, COPY, RUN and CMD. |
| 183 | Which command builds an image named myapp from the current directory? | C) docker build -t myapp . | C) docker build -t myapp . | ✓ | The final dot is the build context. |
| 184 | What does docker run -p 8000:80 myapp do? | A) Maps container port 8000 to host port 80 | D) Maps host port 8000 to container port 80 | ✗ | Format is -p HOST:CONTAINER. |
| 185 | Which command lists running containers? | C) docker ps | C) docker ps | ✓ | Use docker ps -a to include stopped containers. |
| 186 | What is the relationship between an image and a container? | B) An image is a read-only template; a container is a running instance of it | B) An image is a read-only template; a container is a running instance of it | ✓ | Many containers can run from one image. |
| 187 | The CMD instruction in a Dockerfile: | C) Sets the default command run when the container starts (can be overridden) | C) Sets the default command run when the container starts (can be overridden) | ✓ | RUN executes at build time; CMD at container start. |
| 188 | The COPY instruction: | C) Copies files from the build context into the image | C) Copies files from the build context into the image | ✓ | E.g., COPY requirements.txt . |
| 189 | Docker Compose is used to: | D) Compress images | B) Define and run multi-container applications with a YAML file | ✗ | E.g., an API, a vector DB and a UI started with docker compose up. |
| 190 | Docker volumes are used to: | B) Persist data beyond the container's lifecycle | B) Persist data beyond the container's lifecycle | ✓ | Data in the container's writable layer is lost when the container is removed. |
| 191 | A common way to reduce Docker image size is: | A) Using multi-stage builds and slim base images | A) Using multi-stage builds and slim base images | ✓ | Build in one stage; copy only the artifacts into a small final image. |
| 192 | A .dockerignore file is used to: | A) Exclude files from the build context | A) Exclude files from the build context | ✓ | Exclude .git, venv, data and secrets to make builds faster and safer. |
| 193 | Docker Hub is: | A) A public registry for storing and sharing container images | A) A public registry for storing and sharing container images | ✓ | Push with docker push username/image:tag. |
| 194 | To make the best use of Docker layer caching, you should: | C) Copy all source code first, then install dependencies | B) Copy and install dependencies before copying frequently-changing source code | ✗ | Then code changes don't force reinstalling dependencies. |
| 195 | Which command shows a container's logs? | B) docker inspect-logs | C) docker logs <container> | ✗ | Add -f to follow logs live. |
| **Kubernetes (K8s)** | | | | | |
| 196 | The smallest deployable unit in Kubernetes is a: | A) Pod | A) Pod | ✓ | A Pod wraps one or more containers sharing network and storage. |
| 197 | A Kubernetes Deployment: | C) Manages ReplicaSets to keep a desired number of pod replicas and handle rolling updates | C) Manages ReplicaSets to keep a desired number of pod replicas and handle rolling updates | ✓ | It provides declarative updates for pods. |
| 198 | A Kubernetes Service provides: | A) A stable network endpoint to access a set of pods | A) A stable network endpoint to access a set of pods | ✓ | Pod IPs change; the Service name/IP stays stable. |

| # | Question | Your answer | Correct answer | | Explanation |
|---|---|---|---|---|---|
| 199 | Which Service type provisions an external cloud load balancer? | B) ExternalStorage | C) LoadBalancer | ✗ | ClusterIP is internal-only; NodePort exposes a port on each node. |
| 200 | Which command creates or updates resources from a YAML file? | C) kubectl apply -f deployment.yaml | C) kubectl apply -f deployment.yaml | ✓ | apply is declarative and idempotent. |
| 201 | Which command lists pods in the current namespace? | B) kubectl get pods | B) kubectl get pods | ✓ | Add -A for all namespaces. |
| 202 | Where should you store an API key used by a pod? | A) In a Kubernetes Secret | A) In a Kubernetes Secret | ✓ | ConfigMaps are for non-sensitive configuration. |
| 203 | Kubernetes namespaces are used for: | B) Logical isolation and organization of resources within a cluster | B) Logical isolation and organization of resources within a cluster | ✓ | E.g., separate dev, staging and prod namespaces. |
| 204 | The Horizontal Pod Autoscaler (HPA): | D) Scales the number of pod replicas based on CPU or other metrics | D) Scales the number of pod replicas based on CPU or other metrics | ✓ | Vertical scaling (more resources per pod) is handled by VPA. |
| 205 | A Kubernetes Ingress is used to: | A) Route external HTTP(S) traffic to services based on hosts and paths | A) Route external HTTP(S) traffic to services based on hosts and paths | ✓ | Requires an Ingress controller such as NGINX. |
| 206 | If a pod's readiness probe fails, Kubernetes will: | D) Stop sending it traffic, without restarting it | D) Stop sending it traffic, without restarting it | ✓ | A failing liveness probe is what triggers a container restart. |
| 207 | Which command shows a pod's logs? | B) docker logs <pod-name> | D) kubectl logs <pod-name> | ✗ | Add -f to stream, or -c to pick a container. |
| 208 | Which control plane component stores the cluster's state? | B) etcd | B) etcd | ✓ | etcd is a distributed key-value store. |
| 209 | Which component assigns pods to nodes? | B) kubelet | A) kube-scheduler | ✗ | It considers resource requests, affinity and taints. |
| 210 | How do you roll back a Deployment to its previous version? | B) docker rollback | A) kubectl rollout undo deployment/<name> | ✗ | Check history with kubectl rollout history. |
| **Git & GitHub** | | | | | |
| 211 | Which command copies a remote repository to your machine? | A) git clone <url> | A) git clone <url> | ✓ | Forking is done on GitHub; cloning copies it locally. |
| 212 | What does git add do? | C) Stages changes for the next commit | C) Stages changes for the next commit | ✓ | Staging lets you choose what goes into a commit. |
| 213 | Which command records staged changes with a message? | C) git commit -m 'message' | C) git commit -m 'message' | ✓ | Commits are stored locally until pushed. |
| 214 | Which command uploads local commits to a remote repository? | D) git push | D) git push | ✓ | E.g., git push origin main |
| 215 | git pull is equivalent to: | C) git fetch followed by git merge | C) git fetch followed by git merge | ✓ | It downloads and integrates remote changes. |
| 216 | A Git branch is: | C) An independent line of development | C) An independent line of development | ✓ | Branches let you work on features without affecting main. |
| 217 | A Pull Request (PR) on GitHub is used to: | C) Propose merging changes and request code review | C) Propose merging changes and request code review | ✓ | PRs can run CI checks before merging. |
| 218 | A merge conflict happens when: | B) The same lines are changed differently in two branches being merged | B) The same lines are changed differently in two branches being merged | ✓ | Resolve manually, then commit. |
| 219 | The .gitignore file is used to: | A) Specify files Git should not track | A) Specify files Git should not track | ✓ | E.g., .env, __pycache__, venv/, large data files. |
| 220 | Which command shows modified, staged and untracked files? | D) git status | D) git status | ✓ | git log shows commit history instead. |
| 221 | GitHub Actions workflows are defined in: | B) YAML files under .github/workflows/ | B) YAML files under .github/workflows/ | ✓ | They run CI/CD on events like push and pull_request. |
| 222 | Where should API keys used by GitHub Actions be stored? | B) Directly in the workflow YAML | A) In GitHub repository Secrets | ✗ | Access them with ${{ secrets.MY_KEY }}. |
| **APIs & FastAPI** | | | | | |
| 223 | REST is: | B) An architectural style using HTTP methods to operate on resources | B) An architectural style using HTTP methods to operate on resources | ✓ | Resources are identified by URLs, e.g., /users/42. |
| 224 | Which HTTP method is used to retrieve data? | D) GET | D) GET | ✓ | GET should not change server state. |
| 225 | Which HTTP method is typically used to create a new resource or submit data? | D) POST | D) POST | ✓ | E.g., POST /predict with input JSON. |

| # | Question | Your answer | Correct answer | | Explanation |
|---|---|---|---|---|---|
| 226 | HTTP status code 404 means: | B) Not Found | B) Not Found | ✓ | The requested resource does not exist. |
| 227 | HTTP status code 401 means: | B) Unauthenticated: valid credentials are missing or invalid | B) Unauthenticated: valid credentials are missing or invalid | ✓ | 403 means authenticated but not allowed. |
| 228 | FastAPI is: | D) A high-performance Python web framework with automatic OpenAPI docs | D) A high-performance Python web framework with automatic OpenAPI docs | ✓ | It is popular for serving ML models. |
| 229 | Where are FastAPI's interactive Swagger docs available by default? | C) /docs | C) /docs | ✓ | ReDoc is available at /redoc. |
| 230 | How do you run a FastAPI app defined as app in main.py during development? | D) python main.py --fastapi | A) uvicorn main:app --reload | ✗ | --reload restarts on code changes. |
| 231 | In FastAPI, Pydantic is used for: | B) Request/response data validation using type hints | B) Request/response data validation using type hints | ✓ | Invalid input returns a 422 error automatically. |
| 232 | Which HTTP method is NOT idempotent? | A) POST | A) POST | ✓ | Repeating a POST can create duplicate resources. |
| 233 | The most common data format for REST API payloads is: | B) JSON | B) JSON | ✓ | Content-Type: application/json |
| 234 | HTTP status code 429 means: | A) Too Many Requests (rate limited) | A) Too Many Requests (rate limited) | ✓ | Common with LLM APIs; handle with retry and exponential backoff. |
| **Deployment (Render)** | | | | | |
| 235 | Render is: | C) A cloud platform (PaaS) to deploy web services, static sites and databases from Git | C) A cloud platform (PaaS) to deploy web services, static sites and databases from Git | ✓ | Connect a GitHub repo and Render builds and deploys it. |
| 236 | For a web service on Render, your app should listen on: | C) 0.0.0.0 and the port in the PORT environment variable | C) 0.0.0.0 and the port in the PORT environment variable | ✓ | Binding to localhost makes the app unreachable from outside. |
| 237 | With auto-deploy enabled on Render: | B) Pushing to the connected branch automatically triggers a new deploy | B) Pushing to the connected branch automatically triggers a new deploy | ✓ | This gives simple continuous deployment. |
| 238 | Where should API keys (e.g., OPENAI_API_KEY) go when deploying on Render? | D) Environment variables in the Render dashboard | D) Environment variables in the Render dashboard | ✓ | Never commit secrets to the repository. |
| 239 | Why is the first request to a free-tier Render service sometimes slow? | C) The instance spins down after inactivity and must cold start | C) The instance spins down after inactivity and must cold start | ✓ | Paid instances stay always on. |
| 240 | A suitable start command for a Streamlit app on Render is: | B) uvicorn app:streamlit | D) streamlit run app.py --server.port $PORT --server.address 0.0.0.0 | ✗ | Bind to Render's PORT and all interfaces. |
| **Jenkins & CI/CD** | | | | | |
| 241 | Jenkins is: | A) An open-source automation server used for CI/CD | A) An open-source automation server used for CI/CD | ✓ | It automates build, test and deploy steps. |
| 242 | A Jenkinsfile is: | C) A file defining the pipeline as code, stored in the repository | C) A file defining the pipeline as code, stored in the repository | ✓ | Pipeline-as-code makes CI versioned and reviewable. |
| 243 | In a declarative Jenkinsfile, the top-level block is: | A) pipeline { } | A) pipeline { } | ✓ | Inside it: agent, stages, post, etc. |
| 244 | The correct nesting in a declarative pipeline is: | C) stages > stage > steps | C) stages > stage > steps | ✓ | Each stage contains steps such as sh 'pytest'. |
| 245 | The best way to trigger a Jenkins build automatically on every GitHub push is: | B) A GitHub webhook | B) A GitHub webhook | ✓ | Webhooks notify Jenkins immediately when code changes. |
| 246 | Continuous Integration (CI) means: | A) Frequently merging code into a shared branch with automated builds and tests | A) Frequently merging code into a shared branch with automated builds and tests | ✓ | CI catches integration bugs early. |
| 247 | Continuous Delivery differs from Continuous Deployment because: | D) Delivery keeps code always deployable but needs manual approval for production; Deployment releases automatically | D) Delivery keeps code always deployable but needs manual approval for production; Deployment releases automatically | ✓ | Both rely on a strong automated pipeline. |

| # | Question | Your answer | Correct answer | | Explanation |
|---|---|---|---|---|---|
| 248 | Jenkins runs on which port by default? | B) 8080 | B) 8080 | ✓ | Access it at http://localhost:8080 |
| 249 | How should secrets like Docker Hub passwords be used in Jenkins pipelines? | A) Store them in Jenkins Credentials and access them with withCredentials | A) Store them in Jenkins Credentials and access them with withCredentials | ✓ | Jenkins masks credential values in logs. |
| 250 | In Jenkins, an 'agent' is: | B) A Git branch | D) A machine or container where pipeline steps execute | ✗ | agent any runs on any available executor. |